

RAJAT SAVDEKAR

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EDUCATION

University of Southern California, Los Angeles

Aug 2024 – May 2026

Master of Science, Computer Science

GPA: 3.6

Coursework: Databases, Cyber Physical Systems, Multimedia Systems

University of Mumbai, Mumbai

Aug 2019 – May 2023

Bachelor of Engineering, Computer Engineering

GPA: 3.7

Coursework: Data Structures, Software Engineering, Computer Networks

EXPERIENCE

Jalgaon Fruit Sales Cooperative Ltd – Agritech Supply Chain Platform

May 2023 – May 2024

Software Engineer

Jalgaon, India

- Reached **100% clerk adoption in one week** for **150 clerks across 5 warehouses** after a card-based UI was abandoned in a day, rebuilding it as a keyboard-only data grid.
- Cut average query time **40%** on high-frequency lookups by decomposing an undocumented legacy monolith and redesigning the **MySQL schema** and indexing, replacing paper logs for 1,000 trucks/day.
- Eliminated **10–15 minute write freezes** from MyISAM table locking with a **strangler** Flask/PostgreSQL write path, holding **zero freezes** through peak harvest at **2x** throughput, and cut deploy time **30%** with Docker CI/CD and pytest.

USC Summer Programs

May 2025 – Jul 2025

Teaching Assistant

Los Angeles, CA

- Led **2 international teams** of 15+ students through Python, Git, and REST API development, guiding them to build and present project websites using Google Sites.

SELECTED PROJECTS & OPEN SOURCE

MockPad – Collaborative Mock-Interview Editor

May 2026

React, Node.js, Yjs, WebSocket, LevelDB, MongoDB

[\[Live\]](#) [\[GitHub\]](#)

- Built a real-time collaborative interview editor on **Yjs CRDTs** over a custom WebSocket layer, reaching **50+ recurring users** on a single **\$6/month** server.
- Diagnosed session freezes to memory growing with **all rooms ever created, not active ones**; cut peak memory **80%** with **zero data loss** by flushing inactive rooms to disk instead of upgrading the server.
- Integrated OpenAI Whisper into a three-timeline **event-sourced session log** (code, audio, captions) for full replay, plus a health dashboard for memory and room counts.

DevStack – Open Source Developer Infrastructure Platform

Mar 2026

TypeScript, Next.js, Node.js, PostgreSQL, Redis, Docker

[\[GitHub\]](#)

- Engineered a durable **workflow engine** on an event-sourced PostgreSQL schema, with TypeScript replay that resumes from activity history **without re-triggering side effects**.
- Designed a self-hostable platform around it spanning feature flags (**sub-5ms** Redis), JWT/RS256 auth with RBAC, and event analytics – deployable via one **Docker Compose** command.
- Built the remaining services – observability and error tracking – behind the same Compose deploy, so a single host runs the full developer stack.

HONORS & PUBLICATIONS

Aura – visionOS Accessibility Assistant. Aug 2025. 2nd place of 24-hour hackathon; built with Swift, SwiftUI, and RealityKit, delivering fully on-device live captions and environmental sound awareness. [\[GitHub\]](#)

Research Publications. Published multiple peer-reviewed papers with **10+ citations** across IEEE and Springer, covering computer vision, image processing, and machine learning for reproducible scientific research. [\[GitHub\]](#)

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL

Backend & APIs: Node.js, Flask, FastAPI, REST APIs, GraphQL, WebSockets, Microservices

AI & Retrieval: LLMs, RAG, LlamaIndex, embeddings, retrieval evaluation, OCR (Tesseract, PaddleOCR), prompt engineering

Data & Messaging: PostgreSQL, MySQL, MongoDB, Redis, event sourcing

Cloud & Infrastructure: AWS, Docker, Docker Compose, CI/CD, Git, Next.js, React